

Section -II

Unit -4 Elements of HTML

Introduction to Elements of HTML -

An **HTML element** is defined by a starting tag. If the element contains other content, it ends with a closing tag, where the element name is preceded by a forward slash as shown below with few tags -

Start Tag	Content	End Tag
<p>	This is paragraph content.	</p>
<h1>	This is heading content.	</h1>
<div>	This is division content.	</div>

So here <p>....</p> is an HTML element, <h1>...</h1> is another HTML element. There are some HTML elements which don't need to be closed, such as <img.../>, <hr /> and
 elements. These are known as **void elements**.

HTML documents consists of a tree of these elements and they specify how HTML documents should be built, and what kind of content should be placed in what part of an HTML document.

Text formatting Tags (Mark up tags)-

HTML provides various formatting tags. They are collectively called as formatting tags because they are meant for the purpose mentioned above. So, in this chapter, you will learn about these formatting tags and how to use them.

Formatting elements in HTML were mainly created for displaying texts of a special type:

Tag	Meanings
<u>	For making text underlined.
	For making the text Bold.

	Used in those texts which are important and needs to highlight.
<i>	For making the text italics.
	For emphasizing the texts.
<mark>	To make the texts marked or highlighted.
<small>	For making texts small in size then the regular texts surrounded.
<big>	For making texts bigger or larger then the rest of the texts surrounded.
	To show texts that have been deleted or removed or replaced.
<sub>	To give a subscript effect to any text.
<sup>	To give a superscript effect to any text.
<strike>	For displaying any text as strikethrough.
	The tag specifies the font face, font size, and color of text.

Let's check this out by using an example program:

Example:

```

<!DOCTYPE html>
<html>
<head>
<title>Formatting elements example</title>
</head>
<body>
  This makes ur<u> text </u> underlined <br/>
  This is a <b> bold </b> man <br/>
  This is a <strong> Strong </strong> creature <br/>
  I stands for <i> italics </i><br/>
  This will <em> emphasize </em> your text <br/>
  This will <mark> highlight </mark> your word <br/>
  This is a <small> small </small>island <br/>
  This is a <big> Big </big> river <br/>
  I want red color, not <del> green </del><br/>
  H<sub> 2 </sub> O <br/>

```



```
x <sup> 3 </sup><br/>
I want red color, not <strike> green </strike><br/>
<font face="Arial" size=3 color="red">Happy New year</font>
</body>
</html>
```

OUTPUT-

This makes ur text underlined

This is a bold man

This is a Strong creature

I stands for *italics*

This will *emphasize* your text

This will highlight your word

This is a small island

This is a Big river

I want red color, not ~~green~~

Please insert THIS instead of THAT

H₂O

x³

I want red color, not ~~green~~

Happy New year

Output:-

Sangola College, Sangola

Lists:-

There are three types of lists as

- Unordered List (Bulleted)
- Ordered List (Numbering)
- Definition List

Unordered List (Bulleted):-

An Unordered lists starts with the tag and ends with . Each list item starts with the tag and ends with . The attribute that can be used in this tag is as

Attribute	Description
TYPE	Specifies the type of the bullet. It takes three possible value as Disk (Fillround), Circle and Square.

Ex:-

```
<UL TYPE=FILLROUND>
  <LI>B.C.S-I</LI>
  <LI>B.C.S-II</LI>
  <LI>B.C.S-III</LI>
</UL>
```

Output:-

- B.C.S-I
- B.C.S-II
- B.C.S-III

2. Ordered List (Numbering):-

An Ordered lists starts with the tag and ends with . Each list item starts with the tag and ends with . The attribute that can be used in this tag is as

Attribute	Description
TYPE	Controls the numbering scheme to be used. TYPE="1" will give numbers i.e. 1, 2, 3 ... TYPE="A" will give Uppercase letters i.e. A, B, C ... TYPE="a" will give Lowercase letters i.e. a, b, c ... TYPE="I" will give Uppercase Roman numbers i.e. I, II, III... TYPE="i" will give Lowercase Roman numbers i.e. i, ii, iii ...
START	Alters the numbering sequence. Can be set to any numbering value.
VALUE	Changes the numbering sequence in the middle of an ordered list. It is to be specified with the tag.

Ex:-

```
<OL TYPE="1" START="1">  
  <LI>B.C.S-I</LI>  
  <LI>B.C.S-II</LI>  
  <LI>B.C.S-III</LI>  
  <LI VALUE="1">B.C.A-I</LI>  
  <LI>B.C.A-II</LI>  
  <LI>B.C.A-III</LI>  
</OL>
```

Output:-

1. B.C.S-I
2. B.C.S-II
3. B.C.S-III

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कडलास नाका, सांगोला
प्रोफा. एन.ए.मुला

1. B.C.A-I
2. B.C.A-II
3. B.C.A-III

3. Definition List:-

A Definition lists values appears within the tags <DL> and </DL>. Definition list consisting of two parts as

Definition Term appears after the tag <DT>
 Definition Description appears after the tag <DD>

Ex:-

```
<DL>
  <DT>Keyboard</DT>
    <DD> An Input Device</DD>
  <DT>Printer</DT>
    <DD> An Output Device</DD>
</DL>
```

Output:-

```
Keyboard
  An Input Device
Printer
  An Output Device
```

* Adding Graphics to HTML Documents:-

HTML allows placing of static and/or animated images in an HTML page. HTML accepts two picture formats .gif and .jpg. These images can be inserted into web page using the tag which takes the name of the image file as an attribute. In addition HTML also allows control of the height, width, border and etc. The tag takes the following attributes:

Attribute	Description
ALIGN	Controls alignment of the text following the image. It takes possible values as TOP, MIDDLE, and BOTTOM. It also controls alignment of the image with respect to VDU screen. It takes possible values as LEFT, CENTER, and RIGHT.
BORDER	Specifies the size of the border to place around the image.
WIDTH	Specifies the width of the image in pixels.
HEIGHT	Specifies the height of the image in pixels.
HSPACE	Indicates the amount of space to the left and right of the image.
VSPACE	Indicates the amount of space to the top and bottom of the image.
SRC	Specifies the location and name of the image file.
ALT	Indicates the text to be displayed in case the browser is unable to display the image specified in SRC attribute.

Ex:-

```
<IMG ALIGN="BOTTOM" BORDER="6" HEIGHT="100" WIDTH="400"
SRC="c:\winxp.gif" ALT="WIN XP LOGO">
```

This is Windows XP.

Output:-



This is Windows XP.

Tables:-

A table is a two dimensional matrix, consisting rows and columns. All table related tags are included in the `<TABLE>...</TABLE>` tags. Each row of the table is described between the tag `<TR>...</TR>`. Each column of the table is described between the tag `<TD>...</TD>` tag.

Table row can be of two types-

1. Header Row: - A row that spans across columns of a table is called Header Row and it is defined by using `<TH>...</TH>` tag. The content of the table header row is automatically centered and in boldface.
2. Data Rows: - individual data cells placed in the horizontal plane creates a data row. These could be single data cell or multiple data cell. Data cells holds data that must be displayed in the table. A data row is defined by using tag `<TD>...</TD>`. By default data row is left justified.

Any special formatting like boldface or italic is done by including appropriate formatting tag inside the `<TR>...</TR>` tags. An image can be placed in a data cell. The attributes with `<TABLE>` tags are as

Attribute	Description
ALIGN	Horizontal alignment is controlled by this attribute. It takes possible values as LEFT , CENTER , and RIGHT .
VALIGN	Controls the vertical alignment of cell contents. It takes possible values as TOP , MIDDLE , and BOTTOM .
BORDER	Specifies the size of the border to place around the table. The border thickness is specified in pixels.
WIDTH	Specifies the width of the table in pixels or percentage of the available screen. If width is not specified, the data cell is adjusted based on the cell contents.

CELLPADDING	This attribute controls the distance between the data in a cell and the boundaries of the cells.
CELLSPACING	Controls the spacing between adjacent cells.
COLSPAN	The colspan attribute used inside a <TH> or <TD> tag. By using this attribute we can merge two or more columns. It takes any integer value.
ROWSPAN	By using this attribute we can merge two or more rows. It takes any integer value.

The Caption:-

Caption can be provided to a table by using the <CAPTION>...</CAPTION> tag. This paired tags appears within the <TABLE>...</TABLE> tags. The table caption can be made to appear above or below the table structure with the help of ALIGN attribute. It takes two possible values as TOP and BOTTOM.

Ex:-

```
<TABLE BORDER="2" WIDTH="40%">
  <CAPTION ALIGN="BOTTOM"> Ex. of RowSpan and Colspan</CAPTION>
  <TR>
    <TH ROWSPAN="2">NAME</TH>
    <TH COLSPAN="3">MARKS</TH>
  </TR>
  <TR>
    <TD>PB</TD>
    <TD>VB</TD>
    <TD>D2K</TD>
  </TR>
  <TR>
    <TD>AJIT</TD>
    <TD>50</TD>
    <TD>45</TD>
    <TD>56</TD>
  </TR>
  <TR>
    <TD>SUNIL</TD>
    <TD>87</TD>
    <TD>67</TD>
    <TD>66</TD>
  </TR></TABLE>
```

सांगोला डोरॉक्स सेंटर
कडलास नाका, सांगोला
प्रोफा. एन.ए.मुला

Output:-

NAME	MARKS		
	PB	VB	D2K
AJIT	50	45	56
SUNIL	87	67	66

Ex. of RowSpan and Colspan

* Linking Documents i.e. Anchor <A>:-

HTML allows linking to other documents as well as image. Clicking on a section of text or an image in one web page will open an entire web page or an image. The text or an image that provides such linking is called *Hypertext* or *Hyperlink* or *Hotspot*. By clicking on the hyperlink navigation to a different web page or image takes place. The browser separates Hyperlinks from normal text. Every Hyperlink

- Appears in blue color
 - The default color setting in a browser for Hyperlink text or image.
 - The color can be set dynamically via an HTML program if required.
- The Hyperlink text/image is underlined.
- When the mouse cursor is placed over it, the standard arrow shape changes to shape of a hand.

To change these link colors, there are three attributes that can be specified with the <BODY> tag as LINK, VLINK, ALINK. Links are created in a web page by using the <A>... tags. Anything written between these tags become a hyperlink or hotspot. By using HREF attribute of the <A> tag the next navigable web page or image can be specified.

Syntax:-

Hyperlink can be of two types:-

- Links to an external document.
- Links (jumps) to a specific place within the same document.

External Document reference:-

We can navigate external document by providing simple providing HREF attribute i.e. giving file name.

Ex:-

```
<A HREF="test.html">Test Web Page</A>
```

By default a hyperlink takes a user to the beginning of the new web page. But at some time it might be necessary to jump a particular location within the new page. To enable a jump to a specified location on a web page, *named anchors* can be set up. Jumping to a particular location on a web page can be summarized in two steps:

Step 1:-

Mark the location to be jumped to i.e. identify the location in a web page to jump to by giving the location a name by using the NAME attribute of <A> tag.

Syntax:-

```
<A NAME = "location_name">
```

Ex:-

```
<A NAME = "First_Point">
```

This identifies a location to be jumped to as First_Point.

Step 2:-

While jumping to a specific web page and specific location on the web page we require name of the web page and name of the location on the web page.

Syntax:-

```
<A HREF="filename.html # location_name"> Any Text </A>
```

Ex:-

```
<A HREF="test.html # First_Point"> My Web Page </A>
```


My web page becomes hotspot and lead to a location named First_Point in the test.html file.

Internal Document reference:-

Sometimes a jump is required to different location in same document. Same two steps required i.e. identify a location with the name and then jump to that location using the location name. The only difference is that the filename.html now will be the current filename i.e. in another word filename.html is not present.

Syntax:-

```
<A NAME="location_name">
```

```
<A NAME="#location_name"> Location Name</A>
```

Ex:-

```
<A NAME="First_Point">
```

```
<A NAME="# First_Point"> My Web Page</A>
```

Ex:-

```
<OL TYPE="1" START="1">
```

```
<A HREF="BCSI.HTM"><LI>B.C.S-I</LI></A>
```

```
<A HREF="BCSII.HTM"><LI>B.C.S-II</LI></A>
```

```
<A HREF="BCSIII.HTM"><LI>B.C.S-III</LI></A>
```

```
<A HREF="BCAI.HTM"><LI VALUE="1">B.C.A-I</LI></A>
```

```
<A HREF="BCAII.HTM"><LI>B.C.A-II</LI></A>
```

```
<A HREF="BCAIII.HTM"><LI>B.C.A-III</LI></A>
```

```
</OL>
```

```
<A HREF="TEST.HTML"><IMG SRC="c:\winxp.gif"></A>
```

Output:-

1. B.C.S-I

2. B.C.S-II

3. B.C.S-III

1. B.C.A-I

`<a name="sample">sample`
Other contents (full page)
`<a href="#sample">click </`

2. B.C.A-II

3. B.C.A-III

* Image Map:-

When a hyperlink is created on an image, clicking on any part of the image will lead to opening of the document specified in the `<A>` tag. If the image is a large image and there is a need to link multiple documents to the same image, there is a technique that divides the image into multiple sections and allows linking of each section to different document and this technique is known as Image Map. Image Maps can be created and applied to an image so specified portions of the image can be linked to a different file/image.

Linking regions of an image map are called hot regions and each hot region is associated with a document. Creating an image map is a two step process:-

Step 1:-

Create an image map i.e. divide the image into various areas. This is done using the `<MAP>...</MAP>` tags. The `<MAP>` tag takes an attribute NAME which is used to refer in HTML file.

Syntax:-

✓ `<MAP NAME="Map_Name">`

Within the `<MAP>` tag the `<AREA>` tag is specified. This tag defines the regions within the image. The `<AREA>` tag takes some attributes as

Attribute	Description
<u>SHAPE</u>	This attribute is used to set shape in the image. The shape of region can be one of the <u>RECT</u> , <u>CIRCLE</u> , <u>POLYGON</u> and <u>DEFAULT</u> .
<u>COORDS</u>	Each of the above shapes takes different coordinates parameters A <u>RECT</u> (rectangle) takes four coords as x1,y1,x2,y2 A <u>CIRCLE</u> takes three coords as center x, center y and radius A <u>POLYGON</u> takes three or more pairs of coords. A <u>DEFAULT</u> shape does not take any parameter and it indicates the portion of the image not specified under any <u>AREA</u> tag.
<u>HREF</u>	Takes the name of the .html file that is linked to particular area on the image.

Ex:-

<MAP NAME="Test_Map">

<AREA SHAPE="RECT" COORDS="50,40,30,60" HREF="Test.html">

<AREA SHAPE="CIRCLE" COORDS="50,50,10" HREF="Home.html">

</MAP>

Step 2:-

Apply the image map to the particular image. For these purpose the tag takes an attribute called USEMAP that takes the name of the image map as a value and applies the map specification to the image. The value is always preceded by # sign.

Syntax:-

```
<IMG USEMAP="#map_name" SRC="Path">
```

Ex:-

```
<IMG SRC="c:\winxp.gif" USEMAP="Test_Map">
```

Ex:-

```
<BODY>
```

```
<MAP NAME="Test_Map">
```

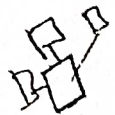
```
<AREA SHAPE="RECT" COORDS="10,10,100,100" HREF="TEST.HTML">
```

```
<AREA SHAPE="CIRCLE" COORDS="100,100,50" HREF="TEST1.HTML">
```

```
</MAP>
```

```
<IMG SRC="c:\winxp.gif" USEMAP="#Test_Map"></A>
```

```
</BODY>
```



* Frames:-

The HTML tags that divide a browser screen into two or more HTML recognizable unique regions is the `<FRAMESET>... </FRAMESET>` tags. Each unique region is called Frame. Each Frame can be loaded with a different document and hence allow multiple HTML documents to be seen concurrently. The HTML frame is powerful feature that enable a web page to broken into different unique section.

The splitting of a browser screen into frames is accomplished with the `<FRAMESET>... </FRAMESET>` tags embedded into HTML document. It requires any one of the following attribute:

Attribute	Description
ROWS	This attribute is used to divide the screen into multiple rows. The values can be a number of Pixels, Percentage of Screen or symbol * which indicates the remaining space.
COLS	This attribute is used to divide the screen into multiple columns. The values can be a number of Pixels, Percentage of Screen or symbol * which indicates the remaining space.

* <FRAME> tag:-

Once the browser screen is divided into rows and columns each unique section defined can be loaded with different HTML documents. This is achieved by using `<FRAME>` tag which takes following attributes:-

Attribute	Description
SRC	Indicates the URL of the document to be loaded into the frame.
MarginHeight="n"	Specify the amount of white space to be left at the top and bottom of the frame.
MarginWidth="n"	Specify the amount of white space to be left along the sides of the frame.
NAME="name"	Gives the unique name to frame. The name given must begin with an alphanumeric character.
NORESIZE	Disable the frames resizing capability.

SCROLLING	Controls the appearance of horizontal and vertical scrolling in a frame. This takes the value YES/NO/AUTO.
-----------	--

The TARGET attribute of <A> tag is used to target the document to particular frame specified in this attribute.

Ex:-

```
<A HREF="Test.html" TARGER="Frame1">
```

Ex:-

```
<html>
```

```
<FRAMESET ROWS="30%,*">
```

```
<FRAME NAME="Frame1" SRC="Title.HTML">
```

```
<FRAMESET COLS="30%,*">
```

```
<FRAME NAME="Frame2" SRC="Link.HTML" NORESIZE SCROLLING="NO">
```

```
<FRAME NAME="Frame3" SRC="Main.HTML" NORESIZE>
```

```
</FRAMESET>
```

```
</FRAMESET>
```

```
</html>
```


What is Multimedia?

Multimedia comes in many different formats. It can be almost anything you can hear or see.

Examples: Images, music, sound, videos, records, films, animations, and more.

Web pages often contain multimedia elements of different types and formats. In this chapter you will learn about the different multimedia formats.

Multimedia Formats

Multimedia elements (like audio or video) are stored in media files.

The most common way to discover the type of a file, is to look at the file extension.

Multimedia files have formats and different extensions like: .swf, .wav, .mp3, .mp4, .mpg, .wmv, and .avi.

HTML5 Audio Tags

Tag	Description	Playing Audio on web browser:-
<audio>	Defines sound content (autoplay, controls, loop, height, width) <i>muted</i>	
<source>	Defines multiple media resources for media elements, such as <video> and <audio> (<i>TYPE, src</i>)	
<!DOCTYPE html> <audio> , <controls> , <type> , height, width, loop,		
<html>		
<body>		
<audio controls>		
<source src="horse.ogg" type="audio/ogg">		
<source src="horse.mp3" type="audio/mpeg">		
Your browser does not support the audio element.		
</audio>		
</body>		
</html>		

Playing video on web browser:-

Tag	Description	<!DOCTYPE html>
<video>	Defines a video or movie (autoplay, controls, loop, height, width) <i>muted</i>	
<source>	Defines multiple media resources for media elements, such as <video> and <audio> (<i>TYPE, src</i>)	
<track>	Defines text tracks in media players	
<body>		
<video width="320" height="240" controls>		
<source src="movie.mp4" type="video/mp4">		


```
<source src="movie.ogg" type="video/ogg">
Your browser does not support the video tag.</video>
</body></html>
```

width	provides the width or breadth of a marquee. For example width="10" or width="20%"
height	provides the height or length of a marquee. For example height="20" or height="30%"
direction	provides the direction or way in which your marquee will allow you to scroll. The value of this attribute can be: left, right, up or down
scrolldelay	provides a feature whose value will be used for delaying among each jump.
scrollamount	provides a value for speeding the marquee feature
behavior	provides the scrolling type in a marquee. That scrolling can be like: sliding, scrolling or alternate
loop	provides how many times the marquee will loop
bgcolor	provides a background color where the value will be either the name of the color or the hexadecimal color-code.
vspace	provides a vertical space and its value can be like: vspace="20" or vspace="30%"
hspace	provides a horizontal space and its value can be like: vspace="20" or vspace="30%"

Marquee Tag:-

The `<marquee>` tag is a container tag of HTML is implemented for creating scrollable text or images within a web page from either left to right or vice versa, or top to bottom or vice versa. But this tag has been deprecated in the new version of HTML, i.e., HTML 5.

The different attributes of `<marquee>` tag are:

Here's are some example of how to use `<marquee>` tag in HTML:

Scroll Up

Example:

```
<marqueewidth="60%"direction="up"height="100px">
```

This is a sample scrolling text that has scrolls in the upper direction.

```
</marquee>
```

Scroll Down

Example:

```
<marqueewidth="60%"direction="down"height="100px">
```

This is a sample scrolling text that has scrolls texts to down.
</marquee>

Scroll Left to Right

Example:

```
<marqueewidth="60%"direction="right"height="100px">  
This is a sample scrolling text that has scrolls texts to right.  
</marquee>
```

Scroll Right to Left

Example:

```
<marqueewidth="60%"direction="left"height="100px">  
This is a sample scrolling text that has scrolls texts to left.  
</marquee>
```

Scrolling Speed

Marquee speed can be changed using the "scrollmount" attribute. For example, if you are using scrollmount = "1" then it sets the marquee to scroll very slowly, and as you increase the "scrollmount," the scrolling speed will also increase.

Example:

```
<marqueebehavior="scroll"direction="up"scrollamount="1">Slow Scrolling</marquee>  
<marqueebehavior="scroll"direction="right"scrollamount="12">Little Fast  
Scrolling</marquee>  
<marqueebehavior="scroll"direction="left"scrollamount="20">Fast  
Scrolling</marquee>  
<marqueebehavior="scroll"direction="right"scrollamount="50">Very Fast  
Scrolling</marquee>
```

Working with Forms and Controls:-

There are different types of form controls that you can use to collect data using HTML form:

- Text Input Controls
- Checkboxes Controls
- Radio Box Controls
- Select Box Controls
- File Select boxes
- Hidden Controls
- Clickable Buttons
- Submit and Reset Button

Text Input Controls

There are three types of text input used on forms:

- **Single-line text input controls** - This control is used for items that require only one line of user input, such as search boxes or names. They are created using HTML `<input>` tag.
- **Password input controls** - This is also a single-line text input but it masks the character as soon as a user enters it. They are also created using HTML `<input>` tag.
- **Multi-line text input controls** - This is used when the user is required to give details that may be longer than a single sentence. Multi-line input controls are created using HTML `<textarea>` tag.

Single-line text input controls

This control is used for items that require only one line of user input, such as search boxes or names. They are created using HTML `<input>` tag.

Example

Here is a basic example of a single-line text input used to take first name and last name:

```
<!DOCTYPE html>
<html>
<head>
<title>Text Input Control</title>
</head>
<body>
<form>
First name: <input type="text" name="first_name" /><br>
Last name: <input type="text" name="last_name" />
</form>
</body>
</html>
```

This will produce following result:

First name:

Last name:

Attributes

Following is the list of attributes for `<input>` tag for creating text field.

Attribute	Description
type	Indicates the type of input control and for text input control it will be set to text .
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
value	This can be used to provide an initial value inside the control.

size	Allows to specify the width of the text-input control in terms of characters.
maxlength	Allows to specify the maximum number of characters a user can enter into the text box.

Password input controls

This is also a single-line text input but it masks the character as soon as a user enters it. They are also created using HTML `<input>` tag but type attribute is set to **password**.

Example

Here is a basic example of a single-line password input used to take user password:

```
<!DOCTYPE html>
<html>
<head>
<title>Password Input Control</title>
</head>
<body>
<form>
User ID : <input type="text" name="user_id" /><br>
Password: <input type="password" name="password" />
</form>
</body>
</html>
```

This will produce following result:

User ID :

Password:

Attributes

Following is the list of attributes for `<input>` tag for creating password field.

Attribute	Description
type	Indicates the type of input control and for password input control it will be set to password .
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
value	This can be used to provide an initial value inside the control.
size	Allows to specify the width of the text-input control in terms of characters.
maxlength	Allows to specify the maximum number of characters a user can enter into the text box.

Multiple-Line Text Input Controls

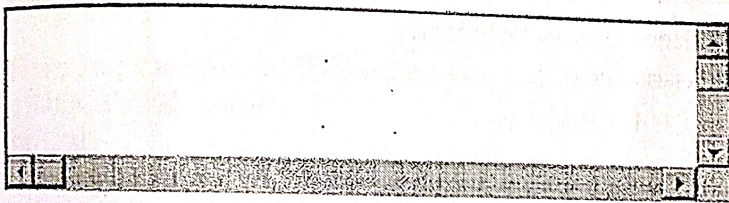
This is used when the user is required to give details that may be longer than a single sentence. Multi-line input controls are created using HTML `<textarea>` tag.

Example

Here is a basic example of a multi-line text input used to take item description:

```
<!DOCTYPE html>
<html>
<head>
<title>Multiple-Line Input Control</title>
</head>
<body>
<form>
Description :<br />
<textarea rows="5" cols="50" name="description">
Enter description here...
</textarea>
</form>
</body>
</html>
```

This will produce following result:
Description :



Attributes

Following is the list of attributes for <textarea> tag.

Attribute	Description
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
rows	Indicates the number of rows of text area box.
cols	Indicates the number of columns of text area box

Checkbox Control

Checkboxes are used when more than one option is required to be selected. They are also created using HTML <input> tag but type attribute is set to **checkbox**.

Example

Here is an example HTML code for a form with two checkboxes:

```
<!DOCTYPE html>
<html>
<head>
<title>Checkbox Control</title>
</head>
<body>
```



```

<form>
<input type="checkbox" name="maths" value="on"> Maths
<input type="checkbox" name="physics" value="on"> Physics
</form>
</body>
</html>

```

This will produce following result:

☐ Maths ☐ Physics

Following is the list of attributes for <checkbox> tag.

Attribute	Description
type	Indicates the type of input control and for checkbox input control it will be set to checkbox .
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
value	The value that will be used if the checkbox is selected.
checked	Set to checked if you want to select it by default.

Radio Button Control

Radio buttons are used when out of many options, just one option is required to be selected. They are also created using HTML <input> tag but type attribute is set to **radio**.

Example

Here is example HTML code for a form with two radio buttons:

```

<!DOCTYPE html>
<html>
<head>
<title>Radio Box Control</title>
</head>
<body>
<form>
<input type="radio" name="subject" value="maths"> Maths
<input type="radio" name="subject" value="physics"> Physics
</form>
</body>
</html>

```

This will produce following result:

☐ Maths ☐ Physics

Attributes

Following is the list of attributes for radio button.

Attribute	Description
type	Indicates the type of input control and for checkbox input control it will be set to radio .
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
value	The value that will be used if the radio box is selected.
checked	Set to checked if you want to select it by default.

Select Box Control

A select box, also called drop down box which provides option to list down various options in the form of drop down list, from where a user can select one or more options.

Example

Here is example HTML code for a form with one drop down box

```
<!DOCTYPE html>
<html>
<head>
<title>Select Box Control</title>
</head>
<body>
<form>
<select name="dropdown">
<option value="Maths" selected>Maths</option>
<option value="Physics">Physics</option>
</select>
</form>
</body>
</html>
```

This will produce following result:

Attributes

Following is the list of important attributes of <select> tag:

Attribute	Description
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
size	This can be used to present a scrolling list box.
multiple	If set to "multiple" then allows a user to select multiple items from the menu.

Following is the list of important attributes of <option> tag:

Attribute	Description
value	The value that will be used if an option in the select box box is selected.
selected	Specifies that this option should be the initially selected value when the page loads.
label	An alternative way of labeling options

File Upload Box

If you want to allow a user to upload a file to your web site, you will need to use a file upload box, also known as a file select box. This is also created using the `<input>` element but type attribute is set to **file**.

Example

Here is example HTML code for a form with one file upload box:

```
<!DOCTYPE html>
<html>
<head>
<title>File Upload Box</title>
</head>
<body>
<form>
<input type="file" name="fileupload" accept="image/*" />
</form>
</body>
</html>
```

This will produce following result:

Attributes

Following is the list of important attributes of file upload box:

Attribute	Description
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
accept	Specifies the types of files that the server accepts.

Button Controls

There are various ways in HTML to create clickable buttons. You can also create a clickable button using `<input>` tag by setting its type attribute to **button**. The type attribute can take the following values:

Type	Description
submit	This creates a button that automatically submits a form.
reset	This creates a button that automatically resets form controls to their initial values.
button	This creates a button that is used to trigger a client-side script when the user clicks that button.

image This creates a clickable button but we can use an image as background of the button.

Example

Here is example HTML code for a form with three types of buttons:

```
<!DOCTYPE html>
<html>
<head>
<title>File Upload Box</title>
</head>
<body>
<form>
<input type="submit" name="submit" value="Submit" />
<input type="reset" name="reset" value="Reset" />
<input type="button" name="ok" value="OK" />
<input type="image" name="imagebutton" src="/html/images/logo.png" />
</form>
</body>
</html>
```

Hidden Form Controls

Hidden form controls are used to hide data inside the page which later on can be pushed to the server. This control hides inside the code and does not appear on the actual page. For example, following hidden form is being used to keep current page number. When a user will click next page then the value of hidden control will be sent to the web server and there it will decide which page has be displayed next based on the passed current page.

Example

Here is example HTML code to show the usage of hidden control:

```
<!DOCTYPE html>
<html>
<head>
<title>File Upload Box</title>
</head>
<body>
<form>
<p>This is page 10</p>
<input type="hidden" name="pagename" value="10"/>
<input type="submit" name="submit" value="Submit"/>
<input type="reset" name="reset" value="Reset"/>
</form>
</body>
</html>
```

O/P This is page 10

HTML5:

HTML5 is the latest and most enhanced version of HTML. Technically, HTML is not a programming language, but rather a markup language. This tutorial gives very good understanding on HTML5.

HTML5 Overview

HTML5 is the next major revision of the HTML standard superseding HTML 4.01, XHTML 1.0, and XHTML 1.1. HTML5 is a standard for structuring and presenting content on the World Wide Web.

HTML5 is a cooperation between the World Wide Web Consortium (W3C) and the Web Hypertext Application Technology Working Group (WHATWG).

The new standard incorporates features like video playback and drag-and-drop that have been previously dependent on third-party browser plug-ins such as Adobe Flash, Microsoft Silverlight, and Google Gears.

Browser Support:

The latest versions of Apple Safari, Google Chrome, Mozilla Firefox, and Opera all support many HTML5 features and Internet Explorer 9.0 will also have support for some HTML5 functionality.

The mobile web browsers that come pre-installed on iPhones, iPads, and Android phones all have excellent support for HTML5.

HTML5 Syntax

The HTML 5 language has a "custom" HTML syntax that is compatible with HTML 4 and XHTML1 documents published on the Web, but is not compatible with the more esoteric SGML features of HTML 4.

HTML 5 does not have the same syntax rules as XHTML where we needed lower case tag names, quoting our attributes, an attribute had to have a value and to close all empty elements. But HTML5 is coming with lots of flexibility and would support the followings:

□ Uppercase tag names. □ Quotes are optional for attributes. □ Attribute values are optional. □ Closing empty elements are optional.

Need of HTML5:

- 1) Cross Browser Support: Your modern, popular browsers all support HTML5 (Chrome, Firefox, Safari IE9 and Opera) and the HTML5 doctype was created so that all browsers, even the really old and annoying ones, er, IE6 can use it. But just because old browsers recognize the doctype that doesn't mean they can use all the new HTML5 tags and goodies. Fortunately, HTML5 is being built to make things easier and more cross browser friendly so in those older IE browsers that don't like the new tags we can just simply add a Javascript shiv that will allow them to use the new elements:
- 2) Game Development: Yup, that is correct, you can develop games using HTML5's <canvas> tag. HTML5 provides a great, mobile friendly way to develop fun, interactive games. If you've built Flash games before, you'll love building HTML5 games.
- 3) Doctype: <!DOCTYPE html>
Yup that's it, that is the doctype, nothing more, nothing less. Pretty simple right? No more cutting and pasting some long unreadable line of code and no more dirty head tags filled with doctype attributes. You can simply and easily type it out and be happy. The really great thing about it though, beyond the simplicity, is that it works in every browser clear back to the dreaded IE6.
- 4) Video and Audio Support: Forget about Flash Player and other third party media players, make your videos and audio truly accessible with the new HTML5 <video> and <audio> tags. Getting your media to play correctly has always been pretty much a nightmare, you had to use the <embed> and <object> tags and assign a huge list of parameters just to get the thing visible and working correctly. Your media tags just become these nasty, huge chunks of confusing code - segments. HTML5's video and audio tags basically treat them as images; <video src="url"/>. But what about all those parameters like height, width and autoplay? No worries my good man, just define those attributes in the tag just like any other HTML element: <video src="url" width="640px" height="380px" autoplay/>.
- 5) Mobile, Mobile, Mobile: Call it a hunch, but I think mobile technology is becoming more popular these days. I know, that is a pretty crazy assumption and some of you are probably thinking — mobile is just a fad... right. Mobile devices are taking over the world. The adoption of mobile devices continues to grow very rapidly and this means that more and more users will be using their mobile browsers to view your web site or application. HTML5 is the most mobile ready tool for developing mobile sites and apps. With Adobe announcing the death of mobile Flash, you will now count on HTML5 to do your mobile web application development.

The DOCTYPE:

DOCTYPEs in older versions of HTML were longer because the HTML language was SGML based and therefore required a reference to a DTD.

HTML 5 authors would use simple syntax to specify DOCTYPE as follows:

<!DOCTYPE html>

Here is the simple syntax for placeholder attribute:

```
<input type="text" name="search" placeholder="search the web"/>
```

This attribute is supported by latest versions of Mozilla, Safari and Chrome browsers only.

The autofocus attribute:

This is a simple one-step pattern, easily programmed in JavaScript at the time of document load, automatically focus one particular form field.

HTML5 introduced a new attribute called autofocus which would be used as follows:

```
<input type="text" name="search" autofocus/>
```

This attribute is supported by latest versions of Mozilla, Safari and Chrome browsers only.

The required attribute:

Now you do not need to have javascript for client side validations like empty text box would never be submitted because HTML5 introduced a new attribute called required which would be used as follows and would insist to have a value:

```
<input type="text" name="search" required/>
```

This attribute is supported by latest versions of Mozilla, Safari and Chrome browsers only.

Graphics in HTML5: —

1) HTML5 – SVG :

SVG stands for Scalable Vector Graphics and it is a language for describing 2D-graphics and graphical applications in XML and the XML is then rendered by an SVG viewer.

SVG is mostly useful for vector type diagrams like Pie charts, Two-dimensional graphs in an X,Y coordinate system etc.

SVG became a W3C Recommendation 14. January 2003 and you can check latest version of SVG specification at SVG Specification. Viewing SVG Files:

Most of the web browsers can display SVG just like they can display PNG, GIF, and JPG.

Internet Explorer users may have to install the Adobe SVG Viewer to be able to view SVG in the browser.

HTML5 - SVG Circle

Following is the HTML5 version of an SVG example which would draw a circle using <circle> tag:

```
<!DOCTYPE html>
```



```

<head> <title>SVG</title>

<meta charset="utf-8" /> </head>

<body>

<h2>HTML5 SVG Circle</h2>

<svg id="svgelem" height="200" xmlns="http://www.w3.org/2000/svg">

  <circle id="redcircle" cx="50" cy="50" r="50" fill="red" />

</svg>

</body>

</html>

```



This would produce following result in HTML5 enabled latest version of Firefox.

HTML5 - SVG Rectangle :

Following is the HTML5 version of an SVG example which would draw a rectangle using

<rect> tag:

```

<!DOCTYPE html>

<head> <title>SVG</title>

<meta charset="utf-8" /> </head>

<body>

<h2>HTML5 SVG Rectangle</h2>

<svg id="svgelem" height="200" xmlns="http://www.w3.org/2000/svg">

  <rect id="redrect" width="300" height="100" fill="red" />

</svg>

</body>

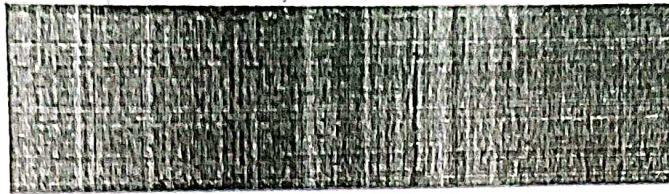
```

<DOCTYPE html>
 <html>
 <body>
 <svg width="300" height="100">
 <circle cx="50" cy="50" r="50">
 stroke="green" stroke-width="2">
 fill="yellow">
 </body></html>



</html>

This would produce following result in HTML5 enabled latest version of Firefox.



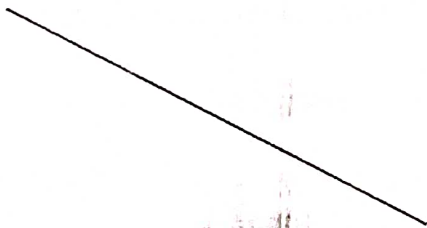
HTML5 - SVG Line

Following is the HTML5 version of an SVG example which would draw a line using <line> tag:

```
<!DOCTYPE html>
<head> <title>SVG</title>
<meta charset="utf-8" /> </head>
<body>
<h2>HTML5 SVG Line</h2>
<svg id="svgelem" height="200" xmlns="http://www.w3.org/2000/svg">
  <line x1="0" y1="0" x2="200" y2="100" style="stroke:red;stroke-width:2"/>
</svg>
</body>
</html>
```

You can use style attribute which allows you to set additional style information like stroke and fill colors, width of the stroke etc.

This would produce following result in HTML5 enabled latest version of Firefox



HTML5 - SVG Ellipse

Following is the HTML5 version of an SVG example which would draw an ellipse using <ellipse> tag:

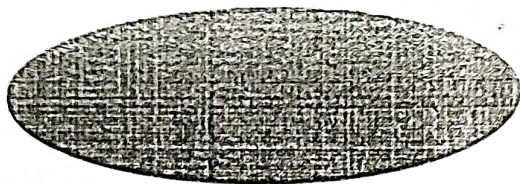
```
<!DOCTYPE html>
<head> <title>SVG</title>
<meta charset="utf-8" />
</head>
<body>
<h2>HTML5 SVG Ellipse</h2>
```



```
<svg id="svgelem" height="200" xmlns="http://www.w3.org/2000/svg">
  <ellipse cx="100" cy="50" rx="100" ry="50" fill="red" />
```

```
</svg>
</body>
</html>
```

This would produce following result in HTML5 enabled latest version of Firefox.



HTML5 - SVG Polygon

Following is the HTML5 version of an SVG example which would draw a polygon using

<polygon> tag:

```
<!DOCTYPE html>
```

```
<head> <title>SVG</title>
```

```
<meta charset="utf-8" />
```

```
</head>
```

```
<body>
```

```
<h2>HTML5 SVG Polygon</h2>
```

```
<svg id="svgelem" height="200" xmlns="http://www.w3.org/2000/svg">
```

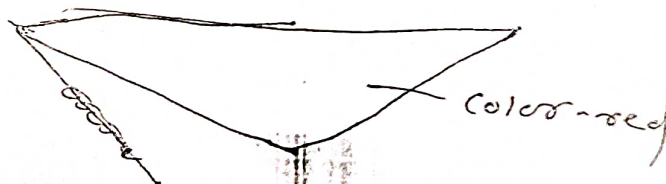
```
  <polygon points="20,10 300,20, 170,50" fill="red" />
```

```
</svg>
```

```
</body>
```

```
</html>
```

This would produce following result in HTML5 enabled latest version of Firefox.



HTML5 - SVG Polyline :

Following is the HTML5 version of an SVG example which would draw a polyline using

<polyline> tag:

```
<!DOCTYPE html>
```

```
<head> <title>SVG</title>
```

```
<meta charset="utf-8" />
```

```
</head>
```

```
<body>
```

```
<h2>HTML5 SVG Polyline</h2>
```